SONG YOUNG OH

(984) 244-9021 | songyoung.oh@duke.edu | www.linkedin.com/in/song-young-oh/ | songyoungoh.github.io

RESEARCH INTERESTS

I am a second-year master's student completing my degree in interdisciplinary data science at Duke University. My research interests lie at the intersection of human-computer interaction and data visualization, exploring how we perceive visual data and translate this perception into actionable insights. I am particularly interested in examining the symbiotic relationship between humans and artificial intelligence, harnessing their combined potential to decode complex datasets and design interactive systems. My aim is to create tools that enrich user-data interactions and empower individuals to make well-informed decisions through seamless human-computer collaboration.

EDUCATION

Duke University, Durham, NC — *August 2022 – Present*

Master of Science in Interdisciplinary Data Science

- Relevant Courses: Modeling and Representation of Data, Data Visualization, Data Engineering Systems, Principles of Machine Learning, Natural Language Processing, Unifying Data Science, Data Science Ethics, Algorithms, Calculus and Probability

Seoul National University, Seoul, South Korea — *February 2015*

Bachelor of Arts in Economics

- Graduated *cum laude*

RESEARCH EXPERIENCE

VisuaLab at UNC Chapel Hill, Chapel Hill, NC — *May 2023 – Present*

Research Assistant (Advisor: Danielle Albers Szafir)

- Collaborating closely with lab researchers to provide substantial support for visualization research projects aimed at understanding how individuals comprehend visual information and improving user interface design, while actively participating in weekly lab meetings and various academic sessions
- Research with Ian Thomas (Ph.D. Student in Computer Science)
    - Conducted extensive research on interactive visualizations to enhance analysts' understanding of semi-supervised machine learning processes, with the goal of reducing practical barriers to adoption and interpretation
        - Executed an empirical study to evaluate the impact of active learning query policies and uncertainty visualizations that enhance transparency in a classifier's decisions on user trust in autonomous classification
    - Currently involved in an ongoing experimental design study to develop an interactive system for exploring imagery latent spaces, addressing existing limitations within the field
- Research with Jade Kandel (Ph.D. Student in Computer Science)
    - Contributed to the development of a visual analytics system designed for monitoring and analyzing the daily movements of individuals living with Parkinson's disease
    - Captured and processed patient body motion data utilizing the Vicon SDK and implemented an immersive environment reconstruction around the patient's body using Unity and HoloLens technology

PUBLICATION

- I. Thomas, **S. Oh**, D. Albers Szafir. "Assessing Trust in Active Learning Systems: Insights from Query Policies and Uncertainty Visualization." Under review for ACM Conference on Intelligent User Interfaces (IUI) 2024.

PROJECT EXPERIENCE

Tracking Climate Change with Satellites and Aritificial Intelligence *August 2023 – Present*

- Leading Duke's Bass Connections team under the mentorship of Kyle Bradbury, an Assistant Research Professor in the Department of Electrical and Computer Engineering, for the development of machine learning models, using computer vision techniques to analyze satellite imagery and other remotely sensed data with the aim of strengthening strategies for climate change mitigation and adaptation
- Currently focusing on assessing the effectiveness of Contrastive Language-Image Pre-training (CLIP) models in a broad spectrum of climate- and enery-related tasks to ascertain the efficacy of our self-supervised approach for remote sensing applications in the context of climate change

Constructing Quantitative Methods for HIV Researchers *Summer 2023*

- Conducted quantitative research at Duke CFAR (Center for AIDS Research) under the guidance of Cliburn Chan, a Professor of Biostatistics & Bioinformatics, and Justin Pollara, an Associate Professor in Surgery, to gain a comprehensive understanding of the associations between individuals' genetic characteristics and their immune responses following HIV vaccination
- Utilized an extensive array of exploratory data analysis and statistical testing techniques, incorporating biological and genomic data

Development of Containerized Microservice for Data Query *December 2022*

- Delveloped a containerized microservice using FastAPI to analyze the impact of global warming on temperature trends in various countries from 1750 to 2015 (Berkeley Earth dataset), which provided user-friendly data querying through Streamlit

PROFESSIONAL EXPERIENCE

Mirae Asset Global Investments, Seoul, South Korea *November 2015 – July 2022*

Data Analyst & Portfolio Manager

- Performed a rigorous analysis of both traditional and alternative data sources, spanning from public economic indicators to high-frequency data, which enabled the ongoing monitoring of global markets and the generation of weekly reports that provided insights into the prevailing financial conditions in key asset markets
- Collaborated seamlessly with the Quantitative Management Division to develop a range of leading indices, apply NLP techniques to parse FOMC minutes and craft a sentiment score, and formulate data-driven asset allocation algorithms, which led to more profitable trading decisions and a significant increase in funding

TECHNICAL SKILLS

- **Programming Languages:** Python, R, JavaScript, C#
- **Visualization:** D3, Tableau, Adobe Illustrator
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn
- **Database & Big Data:** SQL, Docker, AWS Cloud9, Google Cloud
- **Other:** Unity, Emacs, Microsoft Office
- **Certification:** Microsoft Certified Azure AI Fundamentals
- **Languages:** English [Fluent], Korean [Native]

SERVICE TO THE UNIVERSITY

Duke Interdisciplinary Social Innovators (DISI) *Spring 2023*

- Supported a non-profit organization (Partnership Against Domestic Violence) in the creation of a training platform which allows members of the community to schedule speaking engagements and access webinars